

Full-length article

Unequal sanitation in the Global South: infrastructure and governance shortfalls in northern Minas Gerais, Brazil

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ABSTRACT

The universal provision of public utilities, such as basic sanitation, remains a critical challenge in the Global South, where infrastructure disparities and governance failures perpetuate the exclusion of vulnerable populations. This article analyses the disparities in the provision and management of water, sewerage, solid waste and drainage services in the Brazilian semi-arid region, comparing a regional hub (Montes Claros) with 19 peripheral municipalities in northern Minas Gerais. Using microdata from the National Sanitation Information System (SINISA – 2024), the study develops a matrix of institutional compliance and integrated indicators of coverage and inefficiency. The results confirm the persistence of a deep centre-periphery infrastructure divide and identify severe ‘statistical invisibility’ — characterised by an absence of official data — in localities with lower governance capacity. Furthermore, the study highlights the lack of coordination between sanitation policy areas, with a serious historical neglect of stormwater drainage and solid waste management. It is concluded that current funding metrics and the expansion of privatisation in the sector are insufficient without a regulatory framework that addresses territorial inequalities. The study provides recommendations for improving public policies, highlighting the need to review federal coordination mechanisms and subsidies to ensure water and sanitation security in peripheral regions.

1. Introduction

Achieving universal sanitation provision (water, sewerage, waste, drainage) remains a critical challenge for public health and development in the Global South — despite progress, 3.5 billion people lack safe sanitation facilities and >80% of municipal effluents are discharged untreated (Deshpande et al., 2020; Valette and Colon, 2024). In Brazil, the New Legal Framework for Sanitation, Law 14.026/2020 (Brasil, 2020) (which updated Law 11.445/2007 (Brasil, 2007)), set targets, but the current infrastructure remains insufficient and deeply unequal between regions and between urban centres and peripheral areas (Mendes and Alves, 2026). In Minas Gerais, the expansion of private sector participation under Law 14.026/2020 re-frames the universal access of sanitation within a market-driven framework, with significant implications for water and sewerage services (Soares et al., 2025).

Within this context, the Brazilian concept of basic sanitation comprises five pillars (water, sewerage, waste, drainage, governance –

Borges et al., 2022), but these have evolved disproportionately: the historical focus on water supply has marginalised sewerage and, in particular, drainage, leading to a lack of adequate systems in many cities and exposing populations and the environment to recurring impacts (Mendes and Alves, 2026). This asymmetry remains poorly understood in semi-arid regions and medium-sized peripheral municipalities.

Current literature on institutional change in sanitation in the Global South focuses on barriers, but rarely analyses long-term transformations or integrates water, sewerage, waste and drainage. In Brazil, the SNIS (currently transitioning to the National Sanitation Information System — SINISA) shows: water with 93% coverage, yet with 39% losses; sewerage with 61.9% collection and 49.1% treatment (Borges et al., 2022); solid waste still being disposed of in open dumps; and neglected stormwater drainage (Mendes and Alves, 2026). Regional inequalities persist and are deepening – in the northern part of Minas Gerais (semi-arid region), chronic water shortages are linked to arboviruses (Souza et al., 2020), and open dumps contaminate aquifers (70%

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of samples exceeding legal limits). Pahl-Wostl (2015) presents sanitation as a problem of institutional coordination and multi-scale adaptation, not merely one of infrastructure expansion; the National Policy on Solid Waste (NPSW) has failed to eliminate open dumps, and existing studies cover only 4% of them (Cominetti et al., 2025; Morita et al., 2021). In sanitation, efficiency cannot be reduced to economic criteria, as service provision involves social objectives that the market does not automatically resolve (Barbosa et al., 2016; Spronk, 2010).

This article conducts an analysis of 20 municipalities in northern Minas Gerais, using the contrast between the regional hub (Montes Claros) and its surrounding peripheral municipalities, based on microdata from SINISA (2024) and the development of a Regulatory Compliance Index (RCI). Central hypothesis: the universal provision of sanitation in the Brazilian semi-arid region is hindered by an ‘asymmetry of visibility’ – drainage and waste are neglected in favour of water. Three derived hypotheses: (H1) a persistent centre-periphery infrastructure gap, with no spillover of efficiency due to geographical proximity; (H2) the absence of data in official systems as a political indicator of regulatory omission in municipalities with lower institutional capacity (Dickin and Gabrielsson, 2023); (H3) failure to integrate the five pillars, compromising SDGs 6 and 11 (UN Brasil, 2024a, 2024b). Spatial analysis of these indicators allows for a discussion of the need to reformulate the federal pact and socio-environmental literacy as pillars of water governance (Ortiz-Correa et al., 2016).

2. Materials and methods

2.1. Characterisation of the study area

Covering all micro-regions of the northern Mesoregion of Minas

Gerais (IBGE, 2022), the 20 municipalities were selected on the basis of spatial representativeness and administrative relevance, as shown in Fig. 1. The sample includes Montes Claros (the region's main economic and service hub) and 19 small and medium-sized municipalities in its surrounding peripheral municipalities (Bocaiúva, Bonito de Minas, Capitão Enéas, Claro dos Poções, Cônego Marinho, Coração de Jesus, Francisco Sá, Glaucilândia, Januária, Juramento, Juvenília, Lontra, Manga, Mirabela, Miravânia, Montalvânia, Patis, São João da Lagoa and São João do Pacuí). This configuration allows for an analysis of infrastructural disparities within the regional urban hierarchy, using Montes Claros as a local benchmark. The limited number of municipalities enabled an in-depth audit and manual validation of complex variables (e.g. detection of inconsistencies in the records), ensuring the quality of the empirical data.

2.2. Data source and extraction (SINISA 2024)

Primary microdata extracted from the National Sanitation Information System (SINISA), managed by the Ministry of Cities, relating to the base year 2024. Data processing covered the system's five thematic areas: (I) Municipal Management, (II) Water Supply, (III) Sanitary Sewerage, (IV) Solid Waste and (V) Stormwater Drainage. Data extraction and manipulation were carried out in the RStudio environment (R language). Spatial filtering of the sample and the isolation of key indicators were performed using their original codes (e.g. OGM3004 for Sanitation Plan, IN055 for water coverage, IES0001 for sewerage). The Municipal Management axis comprises three structural indicators selected for analysis: the existence of a Basic Sanitation Plan, the presence of a Regulatory Body, and the establishment of a municipal fund. The exclusive adoption of these three criteria is firmly supported by the

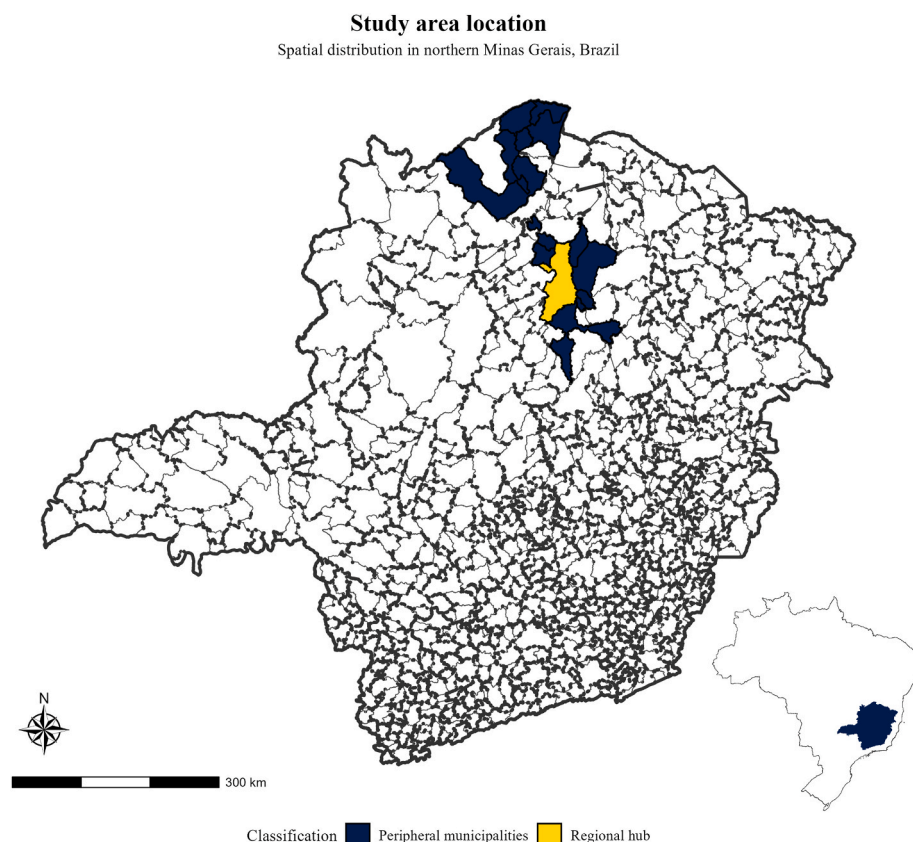


Fig. 1. Geographical location of the study area in northern Minas Gerais, Brazil. Main map illustrates spatial distribution across 20 analysed municipalities, distinguishing Montes Claros as a regional hub (gold) from 19 peripheral towns (navy blue). A thick black outline delineates Minas Gerais state boundaries, while an inset (bottom right) contextualises its position within Brazil. Source: prepared by the authors based on IBGE microdata (2022). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

public policy literature and the Brazilian legal framework (Law No. 11,445/2007 (Brasil, 2007) and Law No. 14,026/2020 (Brasil, 2020)), which enshrine planning, regulation and financing as essential functions of public management.

2.3. Treatment of omissions and ‘statistical invisibility’

Essentially, a central methodological premise of this research was the analytical treatment of data omissions. Values missing from the federal database, reported as Null and/or NA (Not Available) or ‘Not applicable’, were not statistically excluded but actively reclassified as ‘data omission/statistical invisibility’. The study assumes that the persistent lack of responses to the federal system constitutes a failure of governance, highlighting the state's inability to assess the basic conditions for human dignity in peripheral areas. Additionally, a ‘compliance audit’ was conducted by cross-referencing infrastructure and management spreadsheets. For example, nominal sewerage treatment indices were validated against the physical inventory of Wastewater Treatment Plants (WWTPs) to identify administrative inconsistencies (‘ghost infrastructure’).

2.4. Data analysis and visualisation

In order to interpret this dataset, a mixed-methods (quantitative-qualitative) approach was employed for data analysis, with all data processed and visualised entirely within the R environment. Descriptive statistics were generated to compare coverage rates and levels of inefficiency between the regional hub and adjacent municipalities. To map territorial disparities, the *ggplot2* library and its extensions (such as *ggrepel*) were used to generate an institutional compliance matrix (heatmap), scatter plots for the analysis of water supply interruptions, and comparative dashboards.

The Regulatory Normative Compliance Index (RCI) was developed to support the multidimensional synthesis presented in section 3.6 (radar chart grid). As the index incorporates variables of different types, a two-step mathematical process was applied to convert them to a common scale of 0 to 100. Regarding the continuous infrastructure axes (water supply and sanitation), the percentage of population coverage reported for the system was used directly, as the original metric already reflects the proportion of coverage (0 to 100). Finally, for the categorical axes (municipal management, solid waste and stormwater drainage), a weighted binary scoring system was applied, based on the guidelines of the New Legal Framework for Sanitation. Full compliance (e.g. existence of a landfill or regulatory body) received the maximum score, whilst environmental and administrative inadequacy (e.g. operation of open dumps or lack of funding) received a score of zero.

It is crucial to highlight the methodological approach adopted for data omissions (*null/NA*). Rather than being excluded from the sample, omissions were assigned a score of zero in the RCI across all axes. This decision is based on the premise that the index assesses regulatory compliance: the municipality's inability to assess its own territory and report on the service to the Ministry of Cities constitutes a direct violation of the principles of transparency and planning required by law, and is quantified as a complete failure of management.

3. Results and discussion

3.1. Municipal management

Analysis of the regulatory environment reveals that the performance of water and sewerage companies depends on the regulatory environment, and not merely on the method of service delivery (Andres et al., 2024; Carvalho et al., 2023). Without strong institutional alignment, legal reforms risk becoming mere isomorphic mimicry—formally adequate but de facto ineffective (Narzetti and Marques, 2021). An examination of the Municipal Management dimension (SINISA, 2024)

reveals a disparity in institutional capacity between the central municipality and the outlying towns in the region (Fig. 2). Whilst Montes Claros demonstrates formal compliance with the New Legal Framework for Sanitation — attesting to the existence of a Basic Sanitation Plan and a regulatory body for solid waste management — the situation in the adjacent municipalities reveals a serious governance deficit that directly impacts the guarantee of social rights.

The dearth of official data points to a lack of support: 20% of the sample — comprising Montalvânia, São João do Pacuí, Capitão Enéas and Claro dos Poções — showed a complete absence of information (*null/NA*).

Among the municipalities that successfully submitted data to the system, 50% operate urban cleaning services without any regulatory body. Nowhere is this institutional fragility more apparent than in São João da Lagoa, the only municipality to formally confirm the simultaneous absence of long-term sanitation planning, regulatory oversight and earmarked funds.

3.2. Stormwater drainage

Fifty-five per cent of the sample (11 municipalities) provided no information (*null/NA*) regarding the extent of their drainage networks, with localities such as Januária and Claro dos Poções demonstrating an inability to measure their total urban road network. Among the responding municipalities, infrastructure in outlying towns is negligible, with places such as Francisco Sá, Mirabela, and Juramento reporting only around 1 km of underground network for their entire urban area.

Moreover, significant discrepancies exist in the reported data; for instance, Montalvânia recorded a total of merely 0.3 km of urban roads, thereby underscoring the severe technical vulnerability of local administrative frameworks. The absence of consolidated underground networks means, in practice, that surface runoff flows overland, carrying solid waste and untreated sewerage, thereby increasing the risk of disease vectors and flooding.

In general, there is a near-total lack of underground drainage networks in outlying towns, which directly contributes to the occurrence of hydrological disasters and the infringement upon the fundamental rights to adequate housing and life (Fig. 3). The risk management indicators from SINISA (2024) reveal that the burden of state neglect falls exclusively on the population living in low-lying areas.

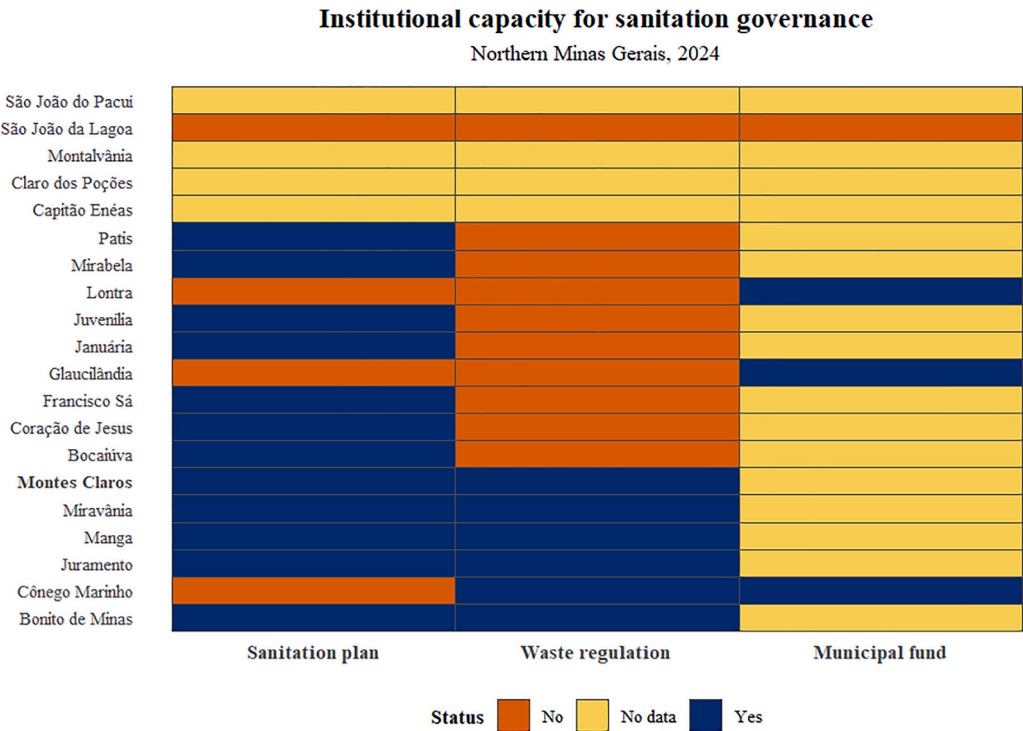
While the main town (Montes Claros) reports no impacts, towns with severe infrastructure deficits have recorded high rates of social vulnerability in the face of the rains. The most extreme case is that of Juvenília, which recorded a ratio of 2.4 displaced/homeless people and a rate of 667 deaths or injuries per 100,000 inhabitants. A similar situation is seen in Bocaiúva, Coração de Jesus and Montalvânia, where housing losses have been recorded.

Municipalities such as Claro dos Poções, Januária, Lontra and São João do Pacuí were unable to monitor and report the impacts of the rains in their areas (*null/NA*).

3.3. Water supply

Urban expansion per se does not ensure universal access to water infrastructure, because coverage depends on the territorial distribution of the systems and not merely on the growth of cities (McDonald et al., 2014). Analysis of water supply indicators reveals that the economic dominance of Montes Claros as a regional hub does not translate into a proportional extension of basic infrastructure to its surrounding peripheral municipalities (Fig. 4). Whilst the hub has a total service coverage rate of 82.7% (IN055), a gradient of exclusion is observed in municipalities such as Bonito de Minas (36.7%) and Cônego Marinho (37.6%), consolidating a coverage gap of over 45 percentage points compared to the regional centre.

Although Montes Claros reports the highest coverage rate (82.7%) alongside operating losses of around 59.6%, the peripheral



Source: SINISA microdata (2024)

Fig. 2. Matrix of compliance with institutional capacity in basic sanitation for municipalities in northern Minas Gerais (2024). The heat map illustrates the regional inequality between the structural adequacy of the hub municipality (Montes Claros) and the administrative vulnerability of peripheral towns in the areas of planning, regulation and financing. Municipalities that failed to report data to the federal system are categorised as ‘No Information’. Source: prepared by the authors based on microdata from SINISA Municipal Management (2024).

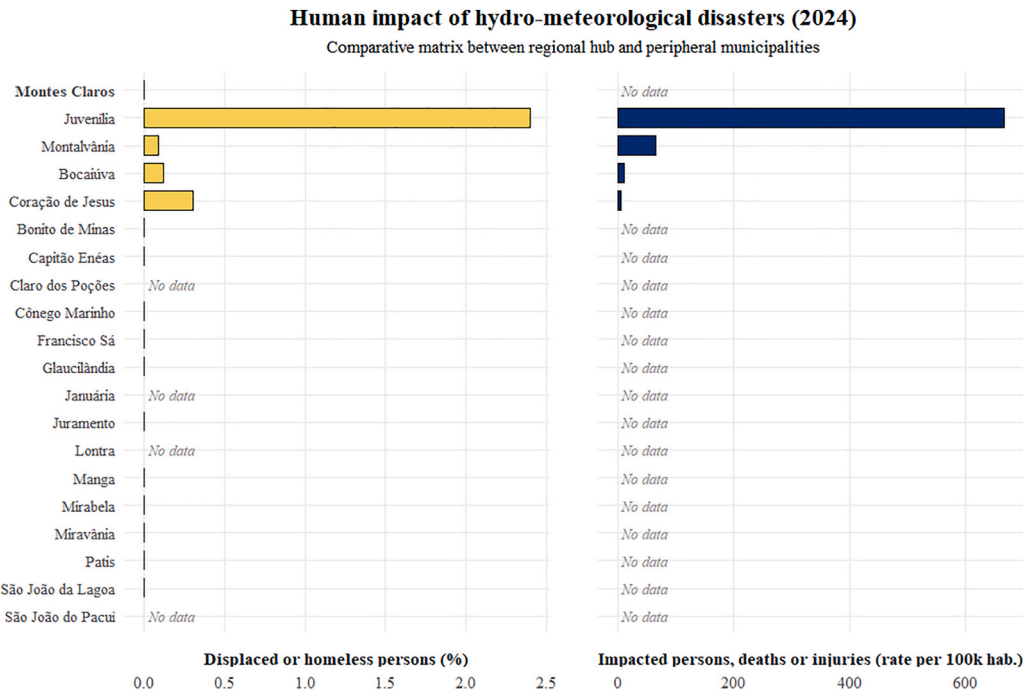


Fig. 3. Human impacts resulting from hydro-meteorological disasters in the municipalities of northern Minas Gerais (2024). The faceted matrix highlights social vulnerability in peripheral localities (notably Juvenília and Bocaiúva), expressed by high rates of homelessness and physical impacts on the population, in contrast to the absence of incidents reported by the central municipality (Montes Claros). Municipalities marked as ‘No data’ (Januária, Claro dos Poções, Lontra and São João do Pacuí) demonstrated a diagnostic inability to monitor risks. Source: prepared by the authors based on SINISA microdata (2024).

municipalities furthest from the hub — such as Bonito de Minas and Cônego Marinho — exhibit severely restricted coverage (<40% to 60%)

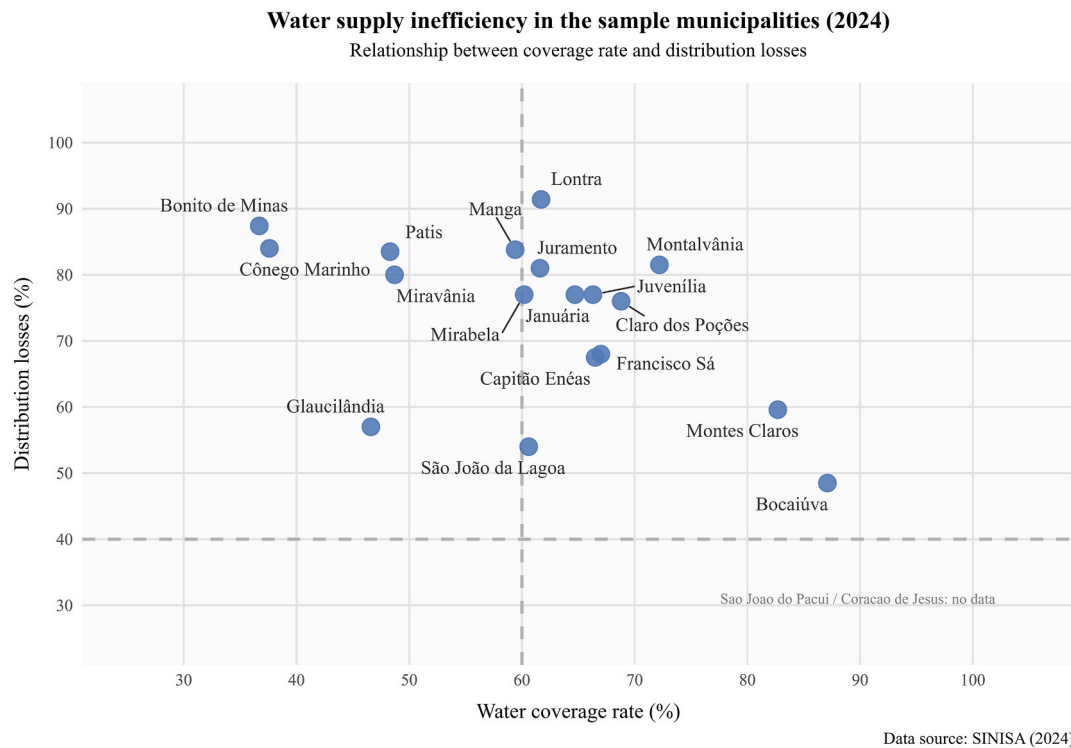


Fig. 4. Inefficiency and exclusion in water supply in northern Minas Gerais (2024). The scatter plot plots the total coverage rate (X-axis) against the distribution loss index (Y-axis). Municipalities located in the top-left quadrant illustrate the dual vulnerability of the periphery: low coverage combined with critical losses exceeding 80%. Source: prepared by the authors based on SINISA microdata (2024).

and losses that soar to around 80% and 90%, indicating that the water reaching the networks is almost entirely wasted before consumption. The clustering of points at the top of the graph demonstrates operational inefficiency due to failures in infrastructure and governance.

Such a landscape of spatial inequality suggests that access to water in northern Minas Gerais is governed by a hierarchy of infrastructure priorities, with smaller municipalities suffering from what might be defined as administrative invisibility. Both the absence of São João do Pacuí from the SINISA 2024 records and the incomplete submission from Coração de Jesus underscore the failure of institutional monitoring; this informational void precludes effective planning for universal access under SDG 6.

Furthermore, detailed analysis reveals that municipalities with more peripheral networks have the highest losses (IN049), as observed in Lontra (91.4%), Bonito de Minas (87.4%) and Manga (83.8%). These figures indicate that the water scarcity problem in the region is caused not only by scarcity but especially by the severe obsolescence of distribution networks and failures in technical management, resulting in the loss of almost all treated water before it reaches the consumer's tap.

3.3.1. Service continuity: widespread instability

Regarding service continuity, an analysis of average total per capita water consumption (IN022), an official indicator compiled by SINISA, reveals a situation that highlights the profound managerial breakdown of the service in the outskirts. Montes Claros, as a regional hub, has an average consumption of 139.2 L/capita/day, placing it at an adequate level of water security and reflecting a more consolidated commercial structure. In contrast, the periphery fluctuates between scarcity and a false surplus. Municipalities such as Cônego Marinho record just 97.5 L/capita/day, operating at the lower limit tolerated by the World Health Organisation (50 to 100 L/capita/day) and highlighting the precarious nature of the water supply. At the opposite extreme, localities such as Glaucilândia show mathematically inflated consumption figures that soar to 261.9 L/capita/day. In the context of the semi-arid region of

Minas Gerais's vulnerability, this peak does not reflect water abundance, but rather attests to severe flaws in water metering, pointing to irregular connections, undetected household leaks, and the structural inability of local providers to diagnose the population's actual consumption.

3.3.2. Administrative and financial management: the geopolitics of investment

As shown in Fig. 5, the allocation of financial resources and operational inefficiencies reflect the same asymmetrical logic. Montes Claros invested R\$ 53.3 million – accounting for 65.3% of the total investment across the 20 municipalities assessed. This figure is 144 times higher than that of São João da Lagoa and seven times that of Januária. However, Montes Claros is no exception to the pattern of structural waste, losing almost 60% of the water it distributes, proving that capital investment alone, without management innovation, does not stem inefficiency.

Taken together, these findings reveal a twofold exclusion – infrastructural (low coverage with critical losses) and statistical (lack of data where services are most precarious). The disparity between the hubs that attract multi-million-dollar budgets and the outlying areas that suffer from water shortages due to operational failures highlights the urgent need for a review of water governance in northern Minas Gerais.

3.4. Sanitation

Investments in water and sanitation reduce hospitalisations due to waterborne diseases, confirming sanitation as a public health infrastructure (Ferreira et al., 2021). In northern Minas Gerais, only two municipalities – Montes Claros (85.2%) and Bocaiúva (77.9%) – have sewerage collection coverage above 70%. Four others (Francisco Sá, Juramento, Capitão Enéas, Glaucilândia) report coverage in the 40–70% range, whilst the majority (nine municipalities) have coverage below 40%, well short of the 90% universalisation target (Fig. 6). Januária (18.9%), Manga (19.2%) and Juvenília (11.1%) are among the lowest.

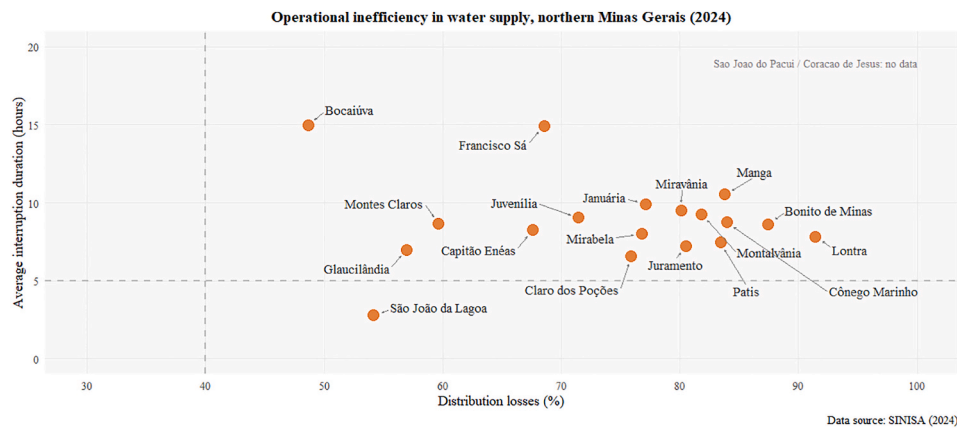


Fig. 5. Impact of operational inefficiency on the continuity of water supply (2024). The relationship between distribution losses (X-axis) and the average duration of interruptions (Y-axis) highlights systemic failures in the region. The dotted lines indicate critical thresholds established for analysis. It is noted that wastage spares neither the regional hub (Montes Claros) nor the municipalities with the most frequent interruptions (such as Bocaiúva and Francisco Sá). Source: prepared by the authors based on SINISA microdata (2024).

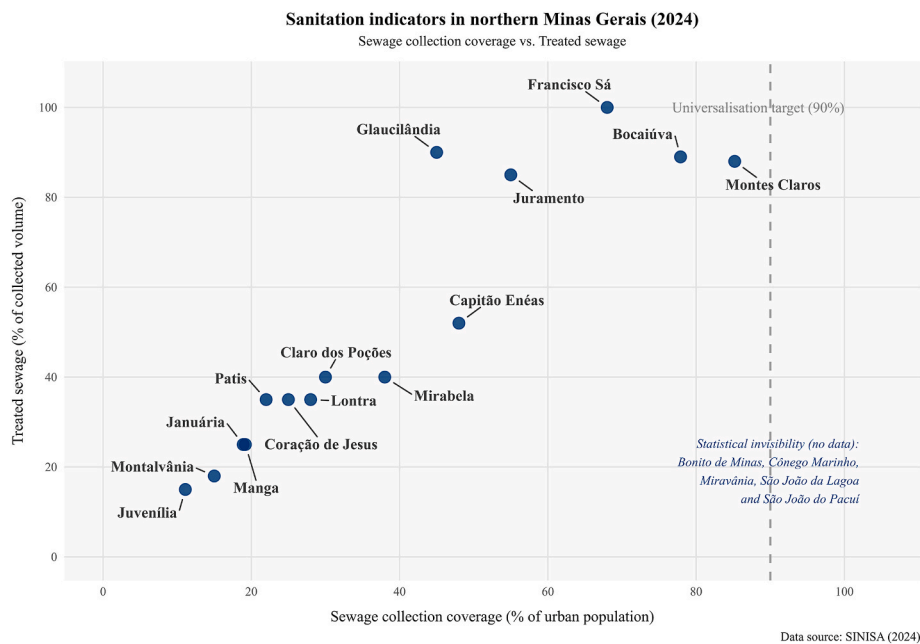


Fig. 6. Sanitation indicators in northern Minas Gerais (2024). The scatter plot illustrates the relationship between sewage collection coverage and the proportion of collected effluent that is treated. The vertical dashed line represents the 90% universalisation target for collection. Municipalities exhibiting 'statistical invisibility' (null/NA data) are explicitly listed to highlight infrastructural and administrative exclusion. Source: prepared by the authors based on SINISA microdata (2024).

Five municipalities (Bonito de Minas, Cônego Marinho, Miravânia, São João da Lagoa, São João do Pacuí) do not report indicators – a statistical invisibility that masks the absence of formal services.

The volume of collected sewerage that actually receives treatment adds a layer of complexity and caution to the analysis. As demonstrated by the clustering in the bottom-left quadrant of Fig. 6, municipalities with already low collection coverage (<20%) also fail to treat the little they collect. In Juvenília, for instance, not only is 88.9% of the sewage left uncollected, but only 15% of the small fraction that is collected is actually treated before returning to the environment.

Municipalities that report universal or near-universal treatment rates (such as Francisco Sá, with a reported 100%) often fail to provide the corresponding register of active Sewerage Treatment Plants (STPs) in SINISA's own physical inventory forms (2024). This discrepancy illustrates the vulnerability of the national system's self-reporting model. Rather than pointing to actual efficiency, these documentary inconsistencies suggest methodological flaws in data entry by local service

providers and a weakness in cross-validation by the federal regulatory body.

In practice, these gaps in traceability undermine the reliability of statistical monitoring, making accurate diagnoses impossible and perpetuating the health deficit among populations living outside the regional hub's investment focus. Overcoming this requires not just funding, but removing institutional barriers to scale up pragmatic, locally adapted wastewater solutions (Reymond et al., 2018).

3.5. Solid waste

Brazil generates around 81 million tonnes of household waste annually; however, almost 41.5% of this volume is still sent to open dumps or unauthorised landfills (Associação Brasileira de Empresas de Limpeza Pública e Resíduos Especiais & Associação Brasileira de Empresas de Limpeza e Resíduos Especiais e Meio Ambiente, 2024), highlighting the persistence of structural deficiencies in solid waste

management and the risks associated with improper disposal (Alfaia et al., 2017), which severely impacts socio-environmental health and deepens the vulnerability of marginalised groups (Gouveia, 2012). In many cases, these peripheral disposal sites function as structural sacrifice zones, exposing populations to the generative dimension of waste and disease governance (Cavalcanti and Lima, 2025). This precarious situation masks profound territorial disparities, as whilst the South and South-East regions have levels of waste collection close to universal coverage, the North and North-East — as well as the outskirts of medium-sized cities — still face significant coverage deficits, where practices such as burning and burying waste remain prevalent (IBGE, 2022; SINISA, 2024). Table 1 illustrates this macro-regional disparity in the provision of waste management services.

Solid waste management highlights the failure of the National Policy on Solid Waste (NPSW — Law 12.305/2010) to eliminate open dumps. In the 2017–2022 series (Fig. 7A), the use of landfill sites remained below 20%; by 2024, only Montes Claros and Capitão Enéas have registered landfill sites. 40% of the sample (eight municipalities) openly use open dumps. Coração de Jesus reports both a rubbish dump and an ‘operating licence’, which contradicts the federal database records.

Juramento and Bocaiúva report household waste collection rates of 100% and 86.5% respectively, but omit data on final disposal (null/NA) – the service is limited to the superficial collection of waste from the urban area, thereby shifting the environmental burden. The financial analysis (SINISA, 2024) demonstrates the unviability of the tariff model under the New Legal Framework: as shown in Fig. 7B and 55% of municipalities do not charge for waste management, whilst 20% omit this information; even if a fee is charged, revenue is negligible – Juramento reported R\$ 3653.00/year (≈US\$ 730).

3.6. Integrated analysis of the axes

The integrated performance of the municipalities was measured by the Regulatory Compliance Index (RCI), which synthesises management and infrastructure indicators on a standardised scale of 0 to 100, resulting in the radar chart presented in Fig. 8. To ensure visual and mathematical comparability between variables of different natures, all axes were converted to represent the percentage of compliance with the ideal target, where 0 indicates a total failure to provide the service (or missing data) and 100 represents full compliance with the guidelines of the New Legal Framework for Sanitation.

For continuous infrastructure services (Water and Sewerage), the percentage of population coverage reported to SINISA was used directly ($RCI_{continuous} = X_{obs}$). Where X_{obs} is the observed coverage rate (0 to 100%).

Regarding the categorical assessment axes (Governance, Solid Waste and Drainage), each regulatory requirement was coded binarily (1 for compliance and 0 for non-compliance or missing data) and aggregated using a weighted average. To eliminate discretion, the assignment of

Table 1

Macro-regional disparities in solid waste management in Brazil. Source: summary of structural indicators based on the Overview of Solid Waste in Brazil (Associação Brasileira de Empresas de Limpeza Pública e Resíduos Especiais & Associação Brasileira de Empresas de Limpeza e Resíduos Especiais e Meio Ambiente, 2024).

Geographical region of Brazil	Proportion of the population covered by regular waste collection (%)	Inappropriate disposal (dumps and controlled landfills) (%)	Inappropriate disposal (dumps and controlled landfills) (%)
Sul	97,2%	> 70,0%	< 30,0%
Sudeste	98,8%	> 65,0%	< 35,0%
Centro-oeste	95,2%	~50,0%	~50,0%
Nordeste	~83,0%	< 40,0%	> 60,0%
Norte	< 80,0%	38,0%	62,0%

weights w_k was anchored in the provisions of the New Legal Framework for Sanitation (Law No. 14,026/2020 (Brasil, 2020), the National Solid Waste Policy (Law No. 12,305/2010 (Brasil, 2010)) and the environmental health guidelines of the World Health Organization (WHO, 2018).

Within the Governance (G) dimension, an equal weighting of 0.5 was assigned to the existence of a municipal basic sanitation plan and 0.5 to affiliation with a Regulatory Agency, given that Brazilian legislation makes the transfer of federal funds conditional upon the joint existence of both instruments. In the Solid Waste (R) axis, final disposal in sanitary landfills was given the highest weighting (0.7) as it constitutes the primary barrier for sanitary and environmental protection, complemented by the existence of institutionalised separate collection (0.3). A similar approach was applied to Drainage (D), prioritising the physical existence of underground network infrastructure (0.7) combined with planning tools, such as the mapping of risk areas (0.3). For any axis j structured by categorical indicators, given that the sum of the assigned weights is equal to 1 ($\sum w_k = 1$), the score is calculated as:

$$RCI_{categorical,j} = \left(\sum_{k=1}^n v_k \cdot w_k \right) \times 100$$

where:

v_k : Binary variable (1 if the municipality meets regulatory requirement k , 0 if it does not meet it or omits data).

w_k : Weight assigned to requirement k within that axis.

n : Total number of requirements considered in the axis.

The relative ranking of the municipalities proved consistent even when subjected to sensitivity analyses using alternative weighting schemes (such as equal or extreme weights). It is essential to emphasise, however, that the RCI acts as a metric of formal compliance. As it is based on indicators attesting to the existence of instruments and infrastructure, the index does not replace the detailed analysis presented in the previous sections and does not, in isolation, assess the actual effectiveness or operational quality of service provision in the daily lives of the population.

4. Discussion

4.1. The asymmetry of visibility and the limits of hypothesis testing

The data extracted from SINISA (2024) reveals a profound disparity in the provision of sanitation services in northern Minas Gerais. However, interpreting these disparities requires caution due to the pervasive ‘statistical invisibility’ observed in the region. As hypothesized (H2), the absence of data in official systems serves not merely as a technical glitch, but as a political indicator of regulatory omission in municipalities with lower institutional capacity. The fact that vulnerable localities such as São João do Pacuí and São João da Lagoa completely failed to report crucial indicators confirms that state neglect begins with a lack of basic informational monitoring.

Because of this severe informational void, our hypotheses regarding a persistent centre-periphery infrastructure gap (H1) and the failure to integrate the five sanitation pillars (H3) cannot be definitively confirmed for the entire region with absolute statistical certainty. Nevertheless, the available data provides compelling evidence supporting these trends. Where data exists, it reveals a clear prioritisation of water supply over drainage and solid waste management, indicating that sanitation services do not evolve in an integrated manner across the region. This pattern is consistent with studies showing that progress in sanitation depends on the coordination of multiple service dimensions and that fragmentation across sectors can hinder broader sustainable development outcomes and municipal planning processes (Carbonell et al., 2023; Mendes and Alves, 2026). Furthermore, the contrast between Montes Claros and its periphery strongly suggests that geographic proximity to a regional hub does not automatically generate an efficiency spillover. Similar territorial disparities in access to sanitation

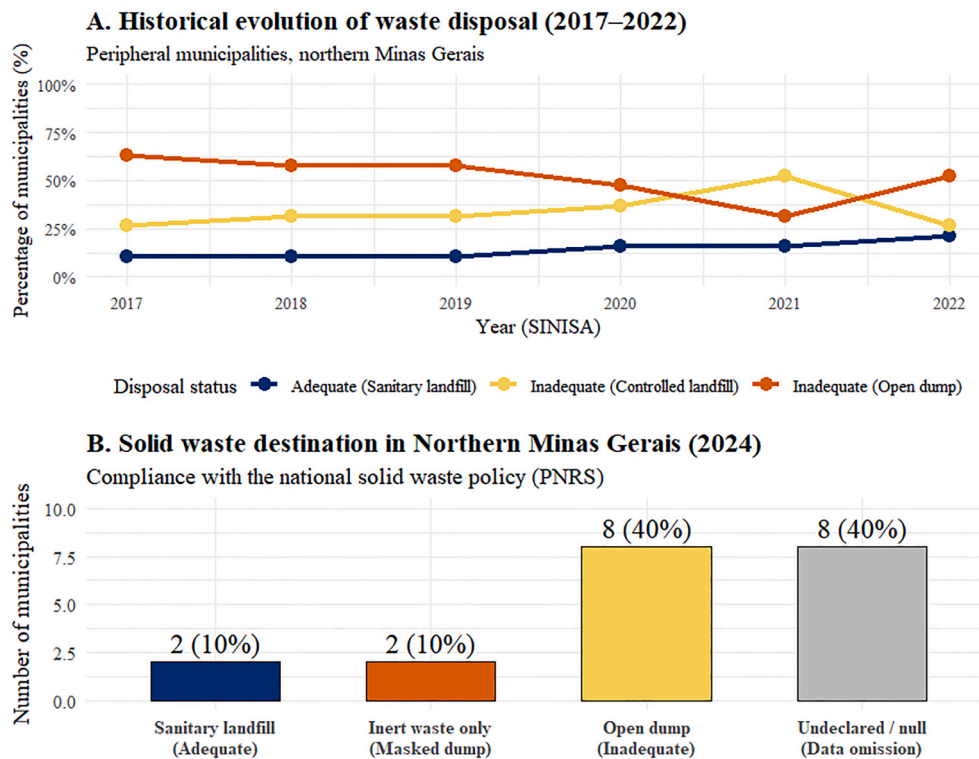


Fig. 7. Final disposal of solid waste in northern Minas Gerais. (A) Historical trends (2017–2022) in peripheral municipalities (excluding Montes Claros), according to SINISA microdata. Categories: Adequate (Sanitary Landfill – navy blue); Inadequate (Controlled Landfill – gold); Inadequate (Open Dump – orange). Percentages total 100% per year. (B) Situation in 2024: only 10% of the sample uses appropriate sanitary landfills; open dumps (gold) persist and data on final disposal is missing (grey). Source: prepared by the authors based on SINISA microdata (2017–2022 and 2024). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

infrastructure have been documented in other developing regions, suggesting that spatial proximity alone is insufficient to guarantee equitable service provision (Deshpande et al., 2020).

4.2. The failure of the privatisation and tariff model in the periphery

The findings demonstrate that investment alone cannot guarantee operational efficiency: despite receiving the majority of regional funding, Montes Claros still loses nearly 60% of its distributed water. Furthermore, relying on user tariffs under the New Legal Framework is financially and socially unsustainable in vulnerable contexts like Juruamento (revenues $\approx$ US\$ 730/year). This model simply shifts the financial burden of universal access onto marginalised households rather than resolving structural coverage deficits (Fagundes et al., 2024, 2025). Consequently, expanding privatisation and financialisation without addressing underlying institutional fragility is not a definitive solution. Such an approach risks the human right to water and sanitation by driving up charges and reinforcing spatial selectivity towards more economically viable municipalities (Britto, 2025; Neves-Silva et al., 2024; Nunes and Marques, 2025). While existing studies note isolated private-sector gains in water supply, there is no consistent evidence that privatisation overcomes territorial inequalities, expands economic accessibility, or systemically improves sewerage and drainage (Pereira et al., 2022; Silva et al., 2024). Ultimately, achieving equitable sanitation depends entirely on the state's capacity to plan, regulate, finance, and integrate policies.

4.3. Policy recommendations for northern Minas Gerais

Based strictly on the empirical evidence gathered from the 20 analysed municipalities, we propose the following targeted policy interventions:

- I) **Regionalised Data Governance:** To overcome the 'statistical invisibility' (H2), the federal and state governments must establish regionalised monitoring protocols. Regional hubs with higher institutional capacity, such as Montes Claros, should be incentivised to provide technical and administrative support to peripheral towns (e.g., Januária, Claro dos Poções) to ensure proper data collection, cross-validation of infrastructure (such as STPs), and consistent reporting to SINISA. Overcoming this barrier is essential, as data insufficiency is a primary driver of inequalities in water, sanitation, and hygiene (WASH) and compromises public policy accountability (Nahas et al., 2023; Sinharoy et al., 2019).
- II) **Targeted Subsidies over Tariff Reliance:** Given the financial incapacity of peripheral municipalities to fund their own solid waste and drainage systems, federal and state funding must pivot towards cross-subsidies and robust social tariffs (Brasil, 2024), which have emerged as key tools for financing sanitation in the Global South (Acey et al., 2019; Narzetti and Marques, 2020). The regionalisation of services should be pursued to create viable economies of scale across the territory (Narzetti et al., 2023), rather than simply to attract private capital to profitable hubs.
- III) **Prioritising Drainage and Waste Integration:** Current funding metrics must be overhauled to require integrated planning. Federal grants for water supply expansion in the region should be made strictly conditional upon simultaneous verifiable investments in stormwater drainage and the eradication of open dumps. Additionally, to address historical infrastructure deficits amidst poor urbanisation and water scarcity, decentralised and nature-based solutions (such as constructed wetlands) should be supported as technically sound, climate-adapted alternatives for semi-arid peripheral territories (Garcia et al., 2022; Lachinani Kordesofla and Abna, 2023; Lemes de Oliveira et al., 2025).



Fig. 8. Radar chart illustrating the Regulatory Compliance Index (RCI) for each municipality (Northern Minas Gerais, 2024). Axes: G (Governance), W (Water), S (Sewerage), R (Solid Waste), D (Drainage). Scores range from 0 at the centre (worst-case scenario) to 100 at the outer edge (ideal compliance). Source: prepared by the authors based on SINISA microdata (2024).

5. Conclusion

This study analysed sanitation inequalities across the municipalities of the Immediate Geographic Region of Montes Claros using SINISA (2024) data and a five-pillar framework encompassing water supply, sewerage, solid waste management, stormwater drainage, and institutional reporting capacity.

The results provide clear evidence of substantial territorial disparities in sanitation infrastructure and governance. Montes Claros concentrates most investments and operational capacity in the region, whereas several peripheral municipalities exhibit severe infrastructure deficits and significant information gaps. However, because the analysis is based on cross-sectional secondary data, the findings do not allow a direct causal assessment of spatial spillover effects. Therefore, H1 should be interpreted as a strong diagnostic indication that proximity to the regional hub has not translated into observable improvements in sanitation performance among neighbouring municipalities, rather than as definitive evidence regarding the mechanisms responsible for this pattern.

Among the three hypotheses, H2 receives the strongest empirical

support. Municipalities with the greatest absence of information in official databases are also those displaying weaker institutional structures and lower monitoring capacity. In this context, missing data should not be viewed merely as a statistical limitation but as an indicator of governance fragility that constrains planning, evaluation, accountability, and access to public investment. The inability to generate and maintain reliable information emerges as a relevant dimension of sanitation inequality itself, corroborating previous discussions on the relationship between information gaps, governance capacity and sanitation inequalities (Dickin and Gabrielsson, 2023; Sinharoy et al., 2019).

The evidence also suggests important fragmentation across sanitation dimensions, lending partial support to H3. The five pillars do not appear to evolve in a coordinated manner, and substantial discrepancies exist among water supply, sewerage, solid waste management, and drainage indicators. Nevertheless, the high incidence of missing values, particularly in drainage data, prevents a rigorous assessment of the causal relationships between sectoral integration and progress towards SDGs 6 and 11. Consequently, the findings should be interpreted as indicating a pattern of institutional and infrastructural fragmentation rather than establishing a definitive statistical relationship. This

interpretation is consistent with previous studies that describe sanitation development as a multidimensional governance challenge requiring coordination across sectors and scales (Pahl-Wostl, 2015; Borges et al., 2022).

A key contribution of this research is demonstrating that data absence constitutes both a methodological challenge and an empirical finding. The uneven reporting capacity observed across municipalities reveals how governance weaknesses can themselves become mechanisms reproducing territorial inequalities. In this sense, the limitations of the database are not external to the phenomenon under investigation; rather, they form part of the governance landscape that shapes sanitation outcomes in semi-arid territories.

Given these constraints, future research should employ longitudinal datasets, field-based validation, and qualitative institutional analyses to investigate the causal pathways linking governance capacity, infrastructure investment, territorial spillovers, and sustainable development outcomes. Such approaches would allow a more robust evaluation of the mechanisms suggested by the present study and provide stronger evidence for policy design in regions characterised by persistent spatial inequalities and persistent sanitation deficits in the Brazilian semi-arid region (Mendes and Alves, 2026; Soares et al., 2025).

CRedit authorship contribution statement

Antônia Valdirene Teixeira Miranda Franco: Writing – original draft, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Millena Miranda Franco:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Iasmim Miranda Franco:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Google Gemini in order to review and standardise the manuscript for British English spelling, grammar, and phrasing. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

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Data availability

Data will be made available on request.

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